Student ID: 24IT045

Chandubhai S Patel Institute of Technology

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
B. TECH (IT/CE/CSE/AIML)

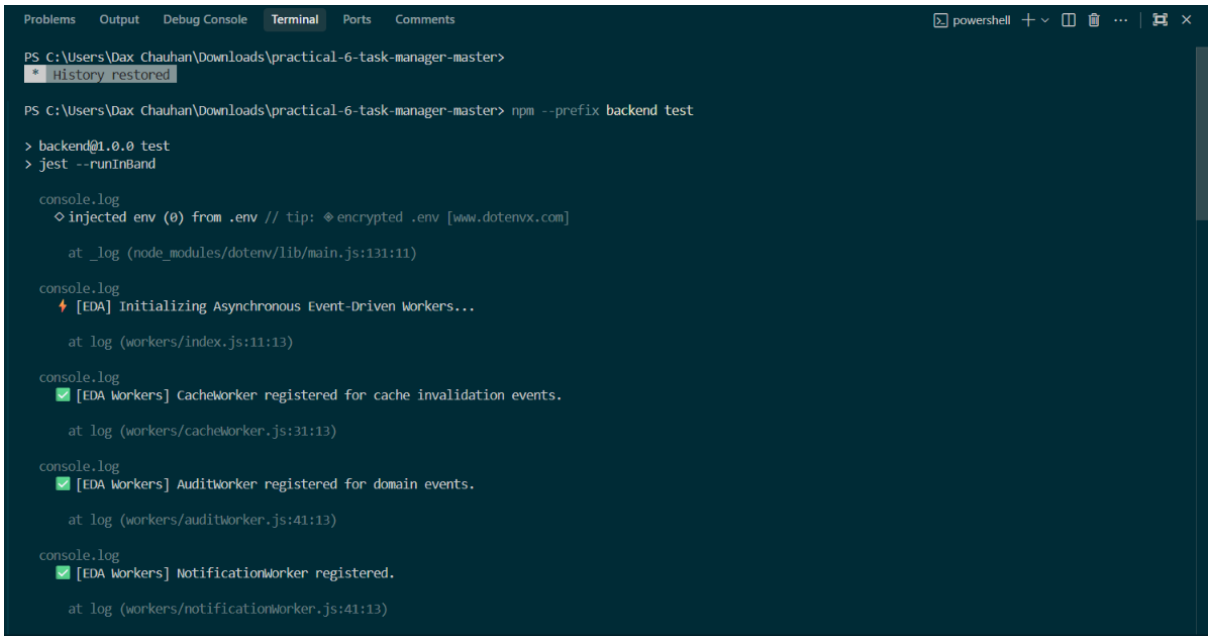
DEPSTAR

Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

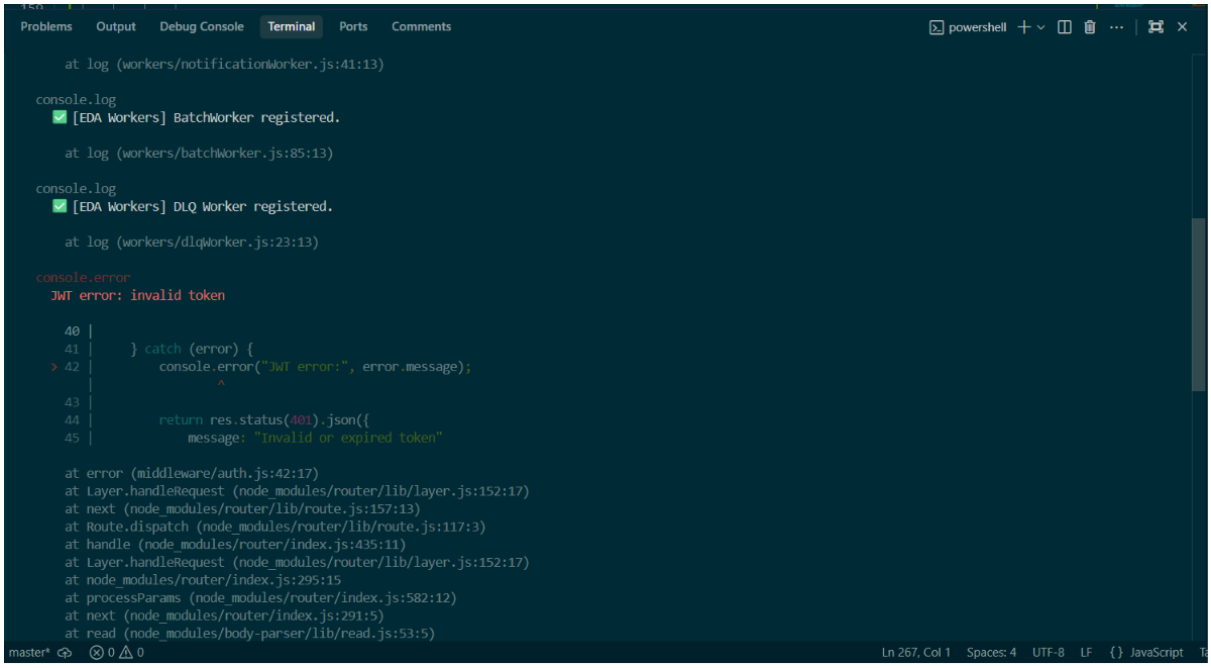
Problem Definition — Practical 13: AI API Integration into a Web Application

To integrate an AI-powered feature into the existing full-stack Task Management application using an external AI service such as OpenAI or Gemini. The system should allow users to generate task descriptions automatically based on task titles and display the generated suggestions in an editable format within the React interface.
The API key must be securely stored on the backend using environment variables and must never be exposed to the frontend. The application should handle API failures gracefully, display loading indicators and appropriate fallback messages, and continue functioning normally when the AI service is unavailable. The implementation should also consider responsible AI usage, data privacy, and user review of AI-generated content.

Screenshots:



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```
Problems  Output  Debug Console  Terminal  Ports  Comments                                  powershell

      at log (workers/notificationWorker.js:41:13)

  console.log
    [EDA Workers] BatchWorker registered.

      at log (workers/batchWorker.js:85:13)

  console.log
    [EDA Workers] DLQ Worker registered.

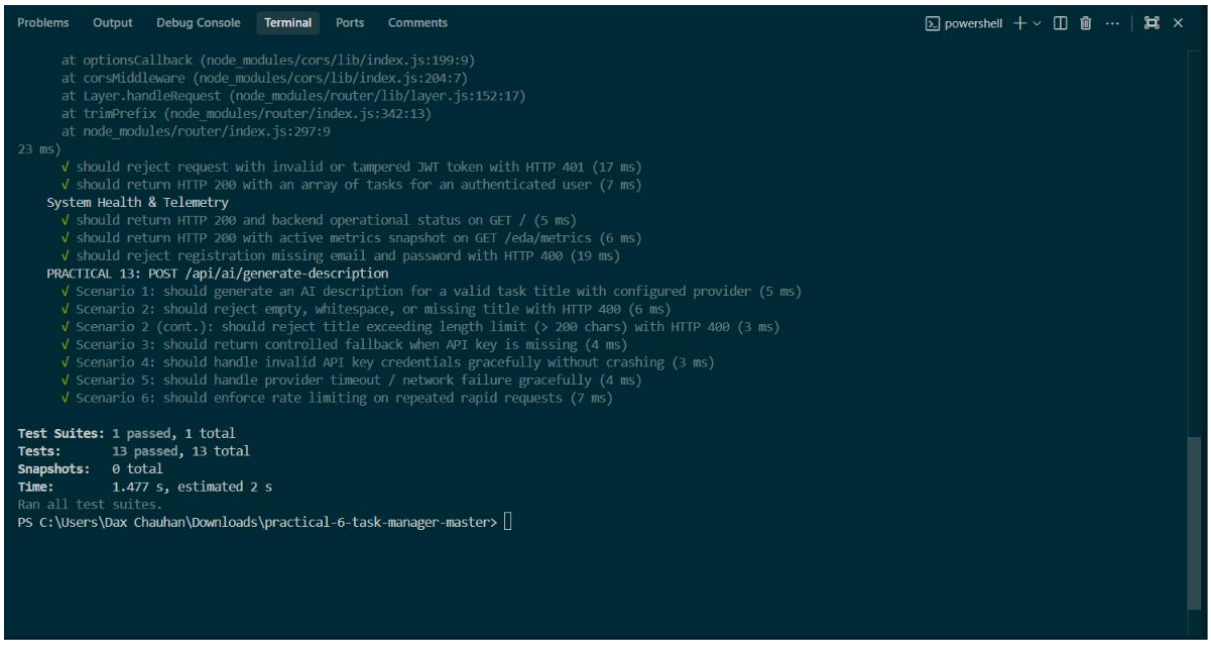
      at log (workers/dlqWorker.js:23:13)

  console.error
    JWT error: invalid token

      40 |
      41 |     } catch (error) {
    > 42 |         console.error("JWT error:", error.message);
         |                 ^
      43 |
      44 |         return res.status(401).json({
      45 |             message: "Invalid or expired token"

      at error (middleware/auth.js:42:17)
      at Layer.handleRequest (node_modules/router/lib/layer.js:152:17)
      at next (node_modules/router/lib/route.js:157:13)
      at Route.dispatch (node_modules/router/lib/route.js:117:3)
      at handle (node_modules/router/index.js:435:11)
      at Layer.handleRequest (node_modules/router/lib/layer.js:152:17)
      at node_modules/router/index.js:295:15
      at processParams (node_modules/router/index.js:582:12)
      at next (node_modules/router/index.js:291:5)
      at read (node_modules/body-parser/lib/read.js:53:5)

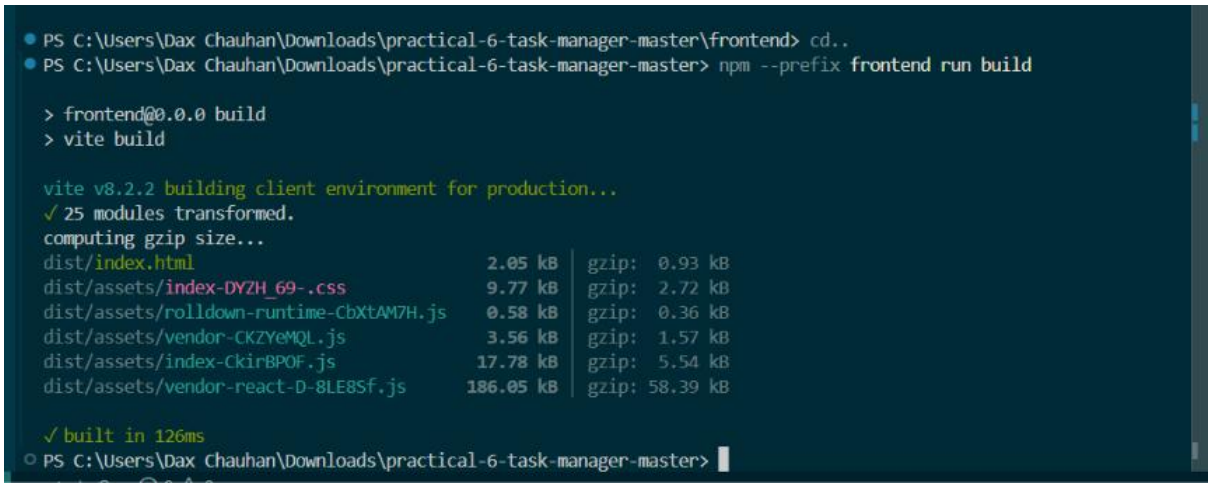
master*  ⊗ 0 ⚠ 0                    Ln 267, Col 1  Spaces: 4  UTF-8  LF  { } JavaScript
```



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Problems  Output  Debug Console  Terminal  Ports  Comments                                  powershell

      at optionsCallback (node_modules/cors/lib/index.js:199:9)
      at corsMiddleware (node_modules/cors/lib/index.js:204:7)
      at Layer.handleRequest (node_modules/router/lib/layer.js:152:17)
      at trimPrefix (node_modules/router/index.js:342:13)
      at node_modules/router/index.js:297:9
23 ms)
      √ should reject request with invalid or tampered JWT token with HTTP 401 (17 ms)
      √ should return HTTP 200 with an array of tasks for an authenticated user (7 ms)
    System Health & Telemetry
      √ should return HTTP 200 and backend operational status on GET / (5 ms)
      √ should return HTTP 200 with active metrics snapshot on GET /eda/metrics (6 ms)
      √ should reject registration missing email and password with HTTP 400 (19 ms)
    PRACTICAL 13: POST /api/ai/generate-description
      √ Scenario 1: should generate an AI description for a valid task title with configured provider (5 ms)
      √ Scenario 2: should reject empty, whitespace, or missing title with HTTP 400 (6 ms)
      √ Scenario 2 (cont.): should reject title exceeding length limit (> 200 chars) with HTTP 400 (3 ms)
      √ Scenario 3: should return controlled fallback when API key is missing (4 ms)
      √ Scenario 4: should handle invalid API key credentials gracefully without crashing (3 ms)
      √ Scenario 5: should handle provider timeout / network failure gracefully (4 ms)
      √ Scenario 6: should enforce rate limiting on repeated rapid requests (7 ms)

Test Suites: 1 passed, 1 total
Tests:       13 passed, 13 total
Snapshots:   0 total
Time:        1.477 s, estimated 2 s
Ran all test suites.
PS C:\Users\Dax Chauhan\Downloads\practical-6-task-manager-master>
```



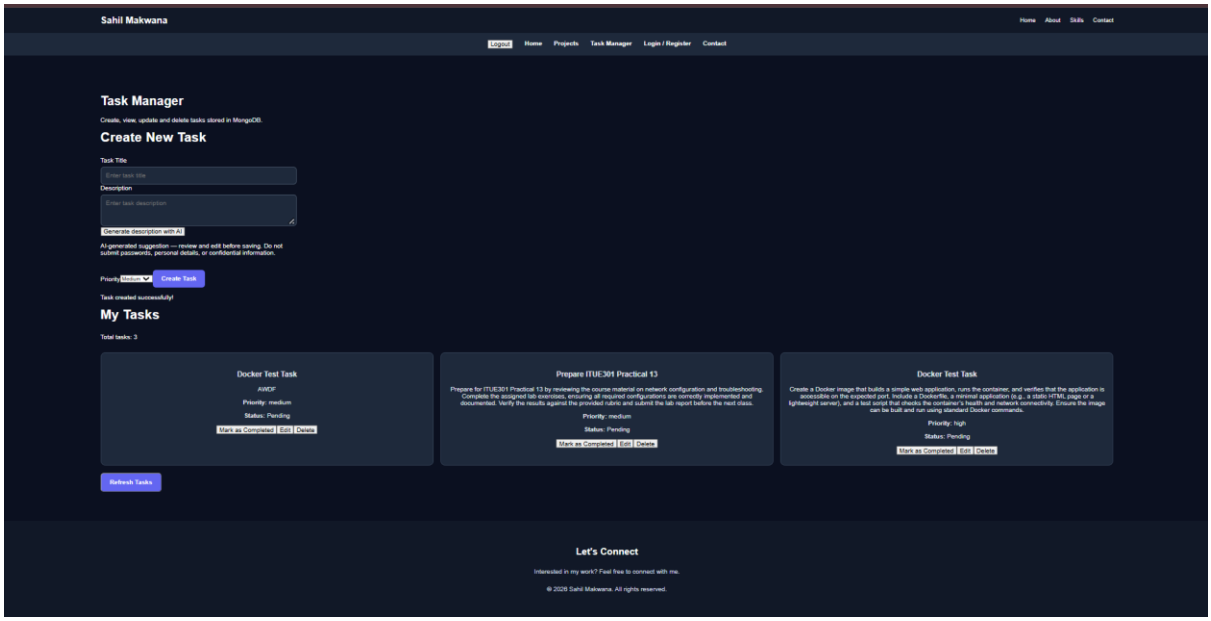
```
PS C:\Users\Dax Chauhan\Downloads\practical-6-task-manager-master\frontend> cd..
PS C:\Users\Dax Chauhan\Downloads\practical-6-task-manager-master> npm --prefix frontend run build

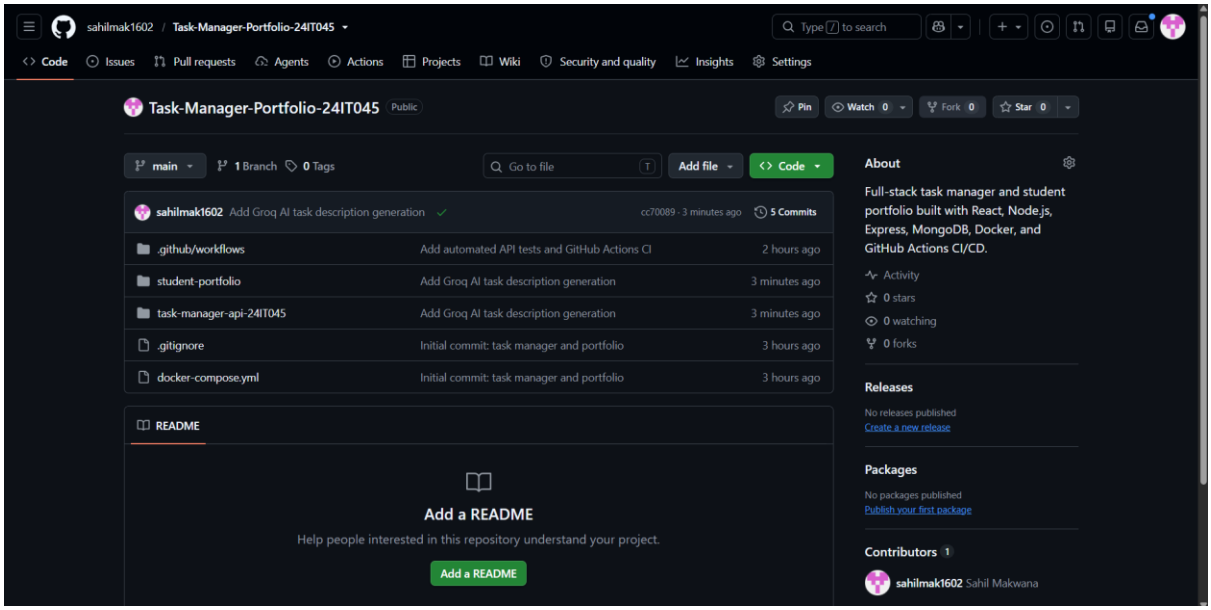
> frontend@0.0.0 build
> vite build

vite v8.2.2 building client environment for production...
√ 25 modules transformed.
computing gzip size...
dist/index.html                   2.05 kB | gzip:  0.93 kB
dist/assets/index-DYZH_69-.css    9.77 kB | gzip:  2.72 kB
dist/assets/rolldown-runtime-CbXtAM7H.js    0.58 kB | gzip:  0.36 kB
dist/assets/vendor-CKZYeMQL.js    3.56 kB | gzip:  1.57 kB
dist/assets/index-CkirBPOF.js    17.78 kB | gzip:  5.54 kB
dist/assets/vendor-react-D-8LE8Sf.js    186.05 kB | gzip: 58.39 kB

√ built in 126ms
PS C:\Users\Dax Chauhan\Downloads\practical-6-task-manager-master>
```

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Github Repo Link: https://github.com/sahilmak1602/Task-Manager-Portfolio-24IT045.git